Answer 1

To find $P(0.5 < x < 5)$ by integrating $f(x)$ from 0.5 to 5 and splitting into three parts: from 0.5 to 1, 1 to 4, and 4 to 5. We get $P(0.5 < x < 5) = 0.1 + 0.3 + 0 = 0.4$.

Answer 2

Remember that the sample space is the set of all possible outcomes. Usually, when you have a random experiment, there are different ways to define the sample space S depending on what you observe as the outcome. In this problem, for each experiment, it is stated what outcomes we observe in order to help you write down the sample space S.

1. We toss a coin until we see two consecutive tails. We record the total number of coin tosses. Here, the total number of coin tosses is a natural number larger than or equal to 2. The sample space is $S = \{2, 3, 4, \ldots\}$.
2. A bag contains 4 balls: one is red, one is blue, one is white, and one is green. We choose two distinct balls and record their colour in order. The sample space can be written as $S = \{(R,B), (B,R), (R,W), (W,R), (R,G), (G,R), (B,W), (W,B), (B,G), (G,B), (W,G), (G,W)\}$.
3. A customer arrives at a bank and waits in line. We observe T. In theory, T can be any real number between 0 and 1/3 = 20 minutes. Thus, S = [0, 1/3] = $\{x \in R \mid 0 \leq x \leq 1/3\}$. .

Answer 3

1. Using the inclusion-exclusion principle, we have
$$P(A \cup B) = P(A) + P(B) - P(A \cap B).$$
Thus,
$$P(A \cap B) = P(A) + P(B) - P(A \cup B) = \frac{1}{2} + \frac{2}{3} - \frac{5}{6} = \frac{1}{3}.$$
2. No, since $A \cap B \neq \phi$.
3. We can write
$$C - (A \cup B) = (C \cup (B \cup C)) - (A \cup B) = S - (A \cup B) = (A \cup B)^c.$$
Thus, $P(C - (A \cup B)) = P((A \cup B)^c) = 1 - P(A \cup B) = 1 - \frac{5}{6} = \frac{1}{6}$.
4. $P(C) = P(C \cap (A \cup B)) + P(C - (A \cup B)) = \frac{5}{12} + \frac{1}{6} = \frac{7}{12}$.

Answer 4

Let A be the event that the first cartridge is defective. Let B be the event that the second cartridge is defective. Let C be the event that the third cartridge is defective. Then the probability that all 3 cartridges are defective is $P(A \cap B \cap C)$.

Hence,

$$P(A \cap B \cap C) = P(A) \cdot P(B|A) \cdot P(C|A \cap B) = \frac{6}{30} \cdot \frac{5}{29} \cdot \frac{4}{28}$$

Answer 5

P$(A \cap B^c) = P(A - (A \cap B)) = P(A) - P(A \cap B) = 0.6 - 0.2,$ Using $P(A^c) = 1 - P(A) = 0.4$

Answer 6

For independent events,

$$P(A \cap B) = P(A) \cdot P(B)$$

as,

$$\begin{aligned} P(A \cup B) &= P(A) + P(B) - P(A \cap B) \\ &= P(A) + P(B) - P(A) \cdot P(B) \\ &= \frac{1}{5} + P - \frac{1}{5}P \\ \Rightarrow \frac{1}{2} &= \frac{1}{5} + \frac{4}{5}P \\ \Rightarrow P &= \frac{3}{8} \end{aligned}$$

Answer 7

$\int_{-\infty}^{\infty} ke^{-x}dx = 1$
$k \cdot \Gamma(1) = 1$
k = 1
Now, E(X) = $\int_{-\infty}^{\infty} xe^{-x}dx = \Gamma(2) = 1$

Answer 8

1. To find $P(X \leq 2, Y \leq 4)$, we can write

   $$P(X \leq 2, Y \leq 4) = P_{XY}(1,2)+P_{XY}(1,4)+P_{XY}(2,2)+P_{XY}(2,4) = \frac{1}{12}+\frac{1}{24}+\frac{1}{6}+\frac{1}{12} = \frac{3}{8}.$$

2. From the table

   $$R_X = \{1, 2, 3\} \text{ and } R_Y = \{2, 4, 5\}$$

   Find the marginal PMFs:

   $$P_X(x) = \begin{cases} \frac{1}{6}, & x = 1 \\ \frac{3}{8}, & x = 2 \\ \frac{11}{24}, & x = 3 \\ 0, & \text{otherwise} \end{cases}$$

   $$P_Y(y) = \begin{cases} \frac{1}{2}, & y = 2 \\ \frac{1}{4}, & y = 4 \\ \frac{1}{4}, & y = 5 \\ 0, & \text{otherwise} \end{cases}$$

3. Using the formula for conditional probability, we have

   $$P(Y = 2|X = 1) = P(X = 1, Y = 2)P(X = 1) = P_{XY}(1,2)P_X(1) = \frac{1}{12}\times\frac{1}{6} = \frac{1}{2}.$$

4. To check whether X and Y are independent, we need to check that

   $P(X = x_i, Y = y_j) = P(X = x_i)P(Y = y_j)$ for all $x_i \in \mathcal{R}_X$ and all $y_j \in \mathcal{R}_Y$.

   Looking at the table and the results from previous parts, we find

   $$P(X = 2, Y = 2) = \frac{1}{6} \neq P(X = 2)P(Y = 2) = \frac{3}{16}.$$

   Thus, we conclude that X and Y are not independent.

Answer 9

To find c, we can use

1. $\int_{-1}^{1} cx^2 dx = 1$
   $\left(\frac{2}{3}\right) C = 1$
   $C = \frac{3}{2}$

2. $E(X) = \int_{-1}^{1} x \left(\frac{3}{2}\right) x^2 dx = 0$
   For finding variance use the formula, $\mathrm{Var}(X) = E(X^2) - [E(X)]^2$.

3. $P(X \geq 1) = \int_{1/2}^{1} \frac{3}{2} x^2 dx = \frac{7}{16}$

Answer 10

$$P(j) \propto \ j$$
$$=kj$$

where k is a constant of proportionality. Next, we determine this constant k by using Theorem P(S)=1, we get
P(S) = P(1) + P(2) + P(3) + P(4) + P(5) + P(6)
= k + 2k + 3k + 4k + 5k + 6k
= (1 + 2 + 3 + 4 + 5 + 6) k
$=\frac{(6)(6+1)}{2}k$

= 21k.

Thus $k = \frac{1}{21}$

Hence, we have P(j) = $\frac{j}{21}$. Now, we want to find the probability of the odd number of dots turning up.
P(odd numbered dot will turn up) = P(1) + P(3) + P(5)

$= \frac{1}{21} + \frac{3}{21} + \frac{5}{21}$

$=\frac{9}{21}$.

Answer 11

Let A be the event of drawing a green marble. The prior probabilities are P(B1) = 13 and P(B2) = 23.

1. The probability that Kathy will win the TV is P(A) = P(A ∩ B1) + P(A ∩ B2)
   $= P(A/B1) \times P(B1) + P(A/B2) \times P(B2)$

   $=\frac{7}{11} \times \frac{1}{3} + \frac{3}{13} \times \frac{2}{3}$

   $=\frac{7}{33} + \frac{2}{13}$

   $=\frac{91}{429} + \frac{66}{429}$

   $=\frac{157}{429}$.
2. Given that Kathy won the TV, the probability that the green marble was selected from B1 is

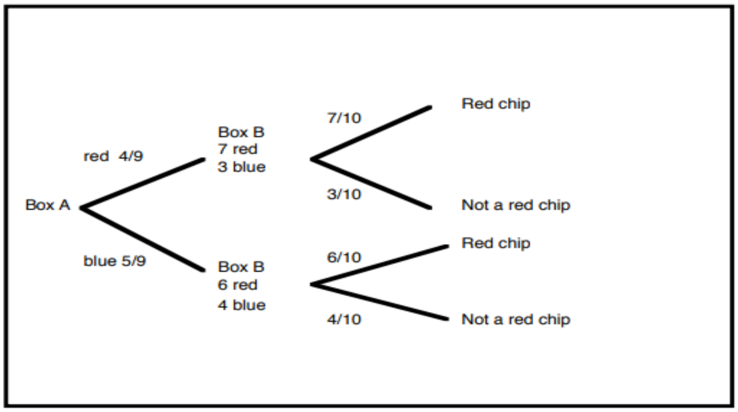

Figure 1: Question 11

P(B1/A) $= \frac{P(A/B1)\times P(B1)}{P(A/B1)\times P(B1)+P(A/B2)\times P(B2)}$

$=\frac{\frac{7}{11}\times\frac{1}{3}}{\frac{7}{11}\times\frac{1}{3}+\frac{3}{13}\times\frac{2}{3}}$

$= \frac{91}{157}$.

Note that P(A/B1) is the probability of selecting a green marble from B1 whereas P(B1/A) is the probability that the green marble was selected from box B1.

Answer 12

Let A denote the event of one who has AIDS and let B denote the event that the test comes out positive.

1. The probability that a person picked at random will test positive is given by
   P(test positive) $= (0.005) \times (0.98) + (0.995) \times (0.03)$
   $= 0.0049 + 0.0298 = 0.035.$

2. The probability that you have AIDS given that your test comes back positive is given by
   P(A/B) $=\frac{\text{favorable positive branches}}{\text{total positive branches}}$

   $=\frac{(0.005)\times(0.98)}{(0.005)\times(0.98)+(0.995)\times(0.03)}$

$=\frac{0.0049}{0.035} = 0.14.$

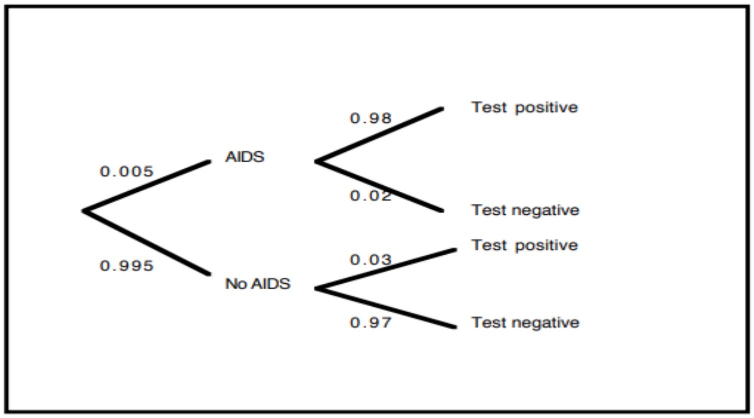

Figure 2: Question 12

Answer 13

Let P and Q be any two points taken at random on the given straight line AB of length a. Let $AP = x$ and $AQ = y$,

$$(0 \leq x \leq a, 0 \leq y \leq a).$$

Then we want $P\{|x - y| > c\}$. The probability can be easily calculated geometrically. Plotting the lines $x - y = c$ and $y - x = c$ along the coordinate axes, we get the following diagram:

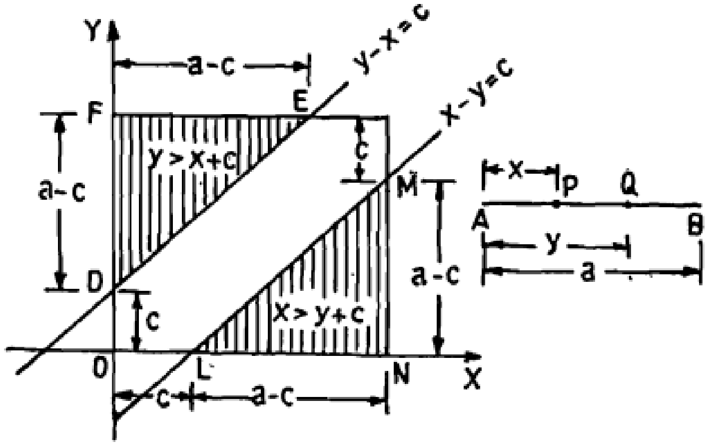

Figure 3: Question 13

Since $0 \leq x \leq a$ and $0 \leq y \leq a$, total area $= a \cdot a = a^2$.
Area favorable to the event $|x-y| > c$ is given by

$$\triangle LMN + \triangle DEF = \frac{1}{2}LN.MN + \frac{1}{2}EF.DF = \frac{(a-c)^2}{2} + \frac{(a-c)^2}{2} = (a-c)^2$$

$$P(|x-y| > c) = \frac{(a-c)^2}{a^2} = \left(1 - \frac{c}{a}\right)^2$$

Answer 14

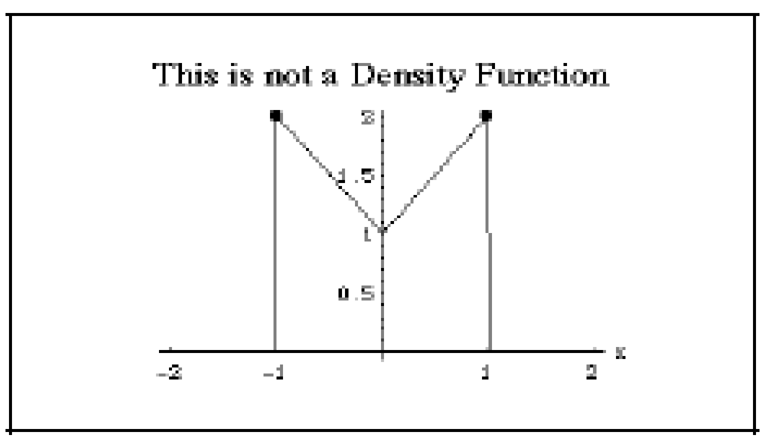

Figure 4: Question 14

It is easy to see that f is nonnegative, that is $f(x) \geq 0$, since

$$f(x) = 1 + |x|$$

. Next, we show that the area under f is not unity. For this, we compute:

$\int_{-1}^{1} f(x)dx = \int_{-1}^{1}(1 + |x|)dx$

$=\int_{-1}^{0}(1 - x)\, dx + \int_{0}^{1}(1 + x)\, dx$

$= \left[x - \frac{1}{2}x^2\right]_{-1}^{0} + \left[x + \frac{1}{2}x^2\right]_{0}^{1}$

$= 1 + \frac{1}{2} + 1 + \frac{1}{2}$

$= 3.$

Thus f is not a probability density function for some random variable X.

Answer 15

The PDF of the random variable is given by

$$f(x) = \frac{d}{dx}F(x) = \frac{d}{dx}\left(\frac{1}{1+e^{-x}}\right) = \frac{d}{dx}\frac{1}{1+e^{-x}}$$

$$= -\frac{1}{(1+e^{-x})^2}\frac{d}{dx}(1+e^{-x}) = -\frac{1}{(1+e^{-x})^2}(-e^{-x})$$

$$= \frac{e^{-x}}{(1+e^{-x})^2}.$$

Answer 16

Note that this density function is symmetric about the y-axis, that is $f(x) = f(-x)$. Hence

$$\int_{-\infty}^{0} f(x)\, dx = \frac{1}{2}.$$

Now we compute the 87.5th percentile q of the above distribution.

$$\begin{aligned}
\frac{87.5}{100} &= \int_{-\infty}^{q} f(x)\,dx \\
&= \int_{-\infty}^{0} \frac{1}{2}e^{-|x|}\,dx + \int_{q}^{0} \frac{1}{2}e^{-|x|}\,dx \\
&= \int_{-\infty}^{0} \frac{1}{2}e^{x}\,dx + \int_{0}^{q} \frac{1}{2}e^{-x}\,dx \\
&= \frac{1}{2} + \int_{0}^{q} \frac{1}{2}e^{-x}\,dx \\
&= \frac{1}{2} + \frac{1}{2} - \frac{1}{2}e^{-q}.
\end{aligned}$$

Therefore, solving for q, we get

$$0.125 = \frac{1}{2}e^{-q} \implies q = -\ln\left(\frac{1}{4}\right) = \ln 4.$$

Answer 17

$$\begin{aligned}
f_1(x) &= \sum_{y=1}^{4} f(x,y) \\
&= \frac{1}{32}\sum_{y=1}^{4}(x+y) \\
&= \frac{1}{32}(4x+10). \\
\text{Therefore, } h\,(y/x) &= \frac{f(x,y)}{f_1(x)} \\
&= \frac{\frac{1}{32}(x+y)}{\frac{1}{32}(4x+10)} \\
&= \frac{x+y}{4x+10}.
\end{aligned}$$

Thus, the conditional probability Y given $X = x$ is

$$h\,(y/x) = \begin{cases} \frac{x+y}{4x+10} & \text{for } x = 1,2; y = 1,2,3,4 \\ 0 & \text{otherwise.} \end{cases}$$

Answer 18

Let G(w) be the cumulative distribution function of W.

$$\begin{aligned}
G(w) &= P(W \le w) \\
&= 1 - P(W > w) \\
&= 1 - P(\min\{X, Y\} > w) \\
&= 1 - P(X > w \text{ and } Y > w) \\
&= 1 - P(X > w)P(Y > w) \text{ (since } X \text{ and } Y \text{ are independent)} \\
&= 1 - \int_w^\infty e^{-x}\,dx \cdot \int_w^\infty e^{-y}\,dy \\
&= 1 - \left(\int_w^\infty e^{-x}\,dx\right)\left(\int_w^\infty e^{-y}\,dy\right) \\
&= 1 - (e^{-w})^2 \\
&= 1 - e^{-2w}.
\end{aligned}$$

Thus, the probability density function of W is

$$g(w) = \frac{d}{dw}G(w) = \frac{d}{dw}\left(1 - e^{-2w}\right) = 2e^{-2w}.$$

Hence

$$g(w) = \begin{cases} 2e^{-2w} & \text{for } w > 0 \\ 0 & \text{elsewhere.} \end{cases}$$

Answer 19

The marginal density of X is given by

$$\begin{aligned}
f_1(x) &= \int_{-\infty}^\infty f(x, y)\,dy \\
&= \int_x^\infty 2e^{-x-y}\,dy \\
&= 2e^{-x}\int_x^\infty e^{-y}\,dy \\
&= 2e^{-x}\left[-e^{-y}\right]_x^\infty \\
&= 2e^{-x}e^{-x} \\
&= 2e^{-2x}.
\end{aligned}$$

Thus, $f_1(x) = 2e^{-2x}$ for $0 < x < \infty$.

Answer 20

$$\begin{aligned} M(t) &= M(0) + \sum_{n=1}^{\infty} M^{(n)}(0)\frac{t^n}{n!} \\ &= M(0) + \sum_{n=1}^{\infty} E(X^n)\frac{t^n}{n!} \\ &= 1 + 0.8\sum_{n=1}^{\infty}\frac{t^n}{n!} \\ &= 0.2 + 0.8 + 0.8\sum_{n=1}^{\infty}\frac{t^n}{n!} \\ &= 0.2e^{0t} + 0.8e^{1t}. \end{aligned}$$

Therefore, we get $f(0) = P(X = 0) = 0.2$ and $f(1) = P(X = 1) = 0.8$. Hence the moment generating function of X is

$$M(t) = 0.2 + 0.8e^t,$$

and the probability density function of X is

$$f(x) = \begin{cases} |x - 0.2| & \text{for } x = 0, 1 \\ 0 & \text{otherwise.} \end{cases}$$

Answer 21

The moment generating function of X is given to be

$$M(t) = \frac{5}{15}e^t + \frac{4}{15}e^{2t} + \frac{3}{15}e^{3t} + \frac{2}{15}e^{4t} + \frac{1}{15}e^{5t}.$$

This suggests that X is a discrete random variable. Since X is a discrete random variable, by definition of the moment-generating function, we see that

$$M(t) = \sum_{x \in R_X} e^{tx} f(x) = e^{tx_1}f(x_1) + e^{tx_2}f(x_2) + e^{tx_3}f(x_3) + e^{tx_4}f(x_4) + e^{tx_5}f(x_5).$$

Hence we have

$$\begin{aligned} f(x_1) &= f(1) = \frac{5}{15}, \\ f(x_2) &= f(2) = \frac{4}{15}, \\ f(x_3) &= f(3) = \frac{3}{15}, \\ f(x_4) &= f(4) = \frac{2}{15}, \end{aligned}$$

$$f(x_5) = f(5) = \frac{1}{15}.$$

Therefore the probability density function of X is given by

$$f(x) = \begin{cases} \frac{6-x}{15}, & \text{for } x = 1, 2, 3, 4, 5, \\ 0, & \text{otherwise.} \end{cases}$$

and the space of the random variable X is $R_X = \{1, 2, 3, 4, 5\}$.

Answer 22

$$\begin{aligned} M_X(t) &= e^{4(e^t-1)} \\ \frac{\partial M}{\partial t} &= e^{4(e^t-1)} \times 4 \times e^t \\ \left.\frac{\partial M}{\partial t}\right|_{t=0} &= M'(0) = 4 \\ E(X) &= 4 \end{aligned}$$

$$\begin{aligned} \frac{\partial^2 M}{\partial t} &= \frac{\partial M}{\partial t}\left(e^{4(e^t-1)} \times 4 \times e^t\right) = 4e^t \times 4e^t \times e^{4(e^t-1)} + 4e^t \times e^{4(e^t-1)} \\ \left.\frac{\partial^2 M}{\partial t}\right|_{t=0} &= M''(0) = 16 + 4 = 20 \\ E(X^2) &= 20 \end{aligned}$$

$$Var(X) = E(X^2) - (E(X))^2 = 4.$$

Answer 23

$$\begin{aligned}
M(s,t) &= e^{(s+3t+2s^2+18t^2+12st)} \\
\frac{\partial M}{\partial s} &= (1+4s+12t)M(s,t) \\
\left.\frac{\partial M}{\partial s}\right|_{(0,0)} &= 1M(0,0) \\
&= 1. \\
\frac{\partial M}{\partial t} &= (3+36t+12s)M(s,t) \\
\left.\frac{\partial M}{\partial t}\right|_{(0,0)} &= 3M(0,0) \\
&= 3. \\
\text{Hence } \mu_X &= 1 \text{ and } \mu_Y = 3.
\end{aligned}$$

Now we compute the product moment of X and Y.

$$\begin{aligned}
\frac{\partial^2 M(s,t)}{\partial s \partial t} &= \frac{\partial}{\partial t}\left(\frac{\partial M}{\partial s}\right) \\
&= \frac{\partial}{\partial t}(M(s,t)(1+4s+12t)) \\
&= (1+4s+12t)\frac{\partial M}{\partial t} + M(s,t)(12). \\
\text{Therefore } \left.\frac{\partial^2 M(s,t)}{\partial s \partial t}\right|_{(0,0)} &= 1(3) + 1(12). \\
\text{Thus} & \\
E(XY) &= 15 \\
\text{and the covariance of } X \text{ and } Y \text{ is given by} & \\
\mathrm{Cov}(X,Y) &= E(XY) - E(X)E(Y) \\
&= 15 - (3)(1) \\
&= 12.
\end{aligned}$$

Answer 24

$$\begin{aligned}
\phi_X(w) &= E(e^{iwx}) \\
&= \int_0^{\infty} \lambda e^{-\lambda x} e^{iwx} \\
&= \left[\frac{\lambda}{iw - \lambda} e^{(iw-\lambda)x} \right]_0^{\infty} \\
&= \frac{\lambda}{\lambda - iw}.
\end{aligned}$$

Note that since $\lambda > 0$, the value of $e^{(iw-\lambda)x}$, when evaluated at x=$+\infty$, is zero.